SIGARAM64

Phase 1 Sprint Proposal

Building the Next Chapter Together

Sprint Window: April 20 - May 30, 2026 • Academic Year Launch: June 2026

From: Professor SuryaKumar S A (Founder- Chess Bishop, FIDE National Instructor)

To: Vishnu T U and Team

1. A Note to the Team

Vishnu and team before we get into scope, timeline, and numbers, I want to start with something that matters more than any of that.

We have worked together for two to three years now. You have been flexible when I needed changes, patient when I asked for additional features, and kind when I delayed payments during rough months. That kind of partnership is not something I take for granted. This document is not a vendor brief - it is a shared roadmap between people who have built something real together.

The coming months are important. We have a Tamil Nadu state-level opportunity that could transform the scale of what Sigaram64 can do for grassroots chess in India. We have 3–4 new districts in the pipeline post-elections. To be ready for all of this, the platform needs a confident upgrade - one that reflects the seriousness of the opportunity without losing the warmth of what we have built.

This Phase 1 proposal is my attempt to structure that upgrade in a way that works for both of us: clear scope, realistic timeline aligned to the academic calendar, reasonable budget, and a shared view of the phases that follow. If anything here does not feel right, let us talk it through.

1.1 Why the Timeline Is What It Is

This proposal gives the team six weeks of build time (April 20 - May 30) instead of a rushed four weeks. The reason is simple: June is when schools reopen for the new academic year across Tamil Nadu. That is when students actually log in and use the platform for the first time. Our real data collection starts from June 15 onwards, after warm-up and stabilisation.

Rushing the build to finish before May 15 serves no purpose when students will not be available until June. Better to use the extra time for quality, polish, and mobile-first testing - so that when students do arrive in June, the platform is solid and impressive.

1.2 What We Are Doing in Phase 1

Phase 1 delivers the vetting-ready foundation. When a senior state-level team evaluates Sigaram64, they should see a modern, credible, measurable platform. That means fixing what is weak (game analysis accuracy, voicebot quality), rebuilding what looks dated (UI), and adding what they will look for (analytics, student management, professional reporting).

We are deliberately keeping this phase tight. The bigger features - Campaign Map, full video lessons, coach dashboard, live classes - come in Phase 2 and Phase 3, which I will discuss gently at the end of this document. Everything is planned. The sequencing just protects both of us: you deliver focused work, I pay on time, and we scale together.

2. Phase 1 Scope

2.1 Core Platform Engineering

Deliverable	Description	Why It Matters
Redesigned UI (Mobile-First & Computer)	Complete UI refresh using React/Next.js. Navy/gold theme matching the Sigaram64 brand. Every screen designed mobile-first then Computer - most government school students access from phones at home.But we Push to use in school system as well. Desktop view adapts responsively.	Government students use phones more than laptops. State vetting teams will test on mobile too. UI must look polished on a 5.5-inch screen.
CAT Adaptive Assessment	8–12 minute assessment with interactive board clicks (no notation typing). Elo range 400–2400. Founder provides content validation If required. Bilingual Tamil/English. Rating import for existing rated players.	Accurately places beginners through FIDE-rated players. Essential for serving the full spectrum we discussed.
Stockfish 18 Game Analysis Fix	Server-side analysis worker at depth 22+, 3-PV, NNUE evaluation. Fixes the current accuracy issues completely.	The current analysis is our biggest credibility risk. Must be bulletproof before any technical vetting.
CPL Move Classification	Brilliant/Best/Good/Inaccuracy/Mistake/Blunder classification with visual icons.	Makes analysis understandable for students at every level.
Glicko-2 Rating System	Replaces basic Elo. Tracks rating deviation and volatility for more accurate ratings.	Industry standard. Used by Lichess. More accurate than basic Elo.
Event Tracking + Analytics Pipeline	Client SDK + Kafka + ClickHouse for logging 20+ interaction types. Foundation for all future reporting.	Without this, we cannot produce the government renewal reports in Phase 2.
Admin Dashboard	Active users, assessment completion rate, puzzle solve rates, per-district filtering.	Gives you and the team visibility into what students are doing day-to-day.
3rd Party Puzzle Engine	Users will be served with tactics from lichess API and some old chess database given by chessbishop.	Justified to their user level based on assessment.
District Activity Management - Manager Login(Staff)	1.The Manager shall upload **GPS-tagged images and videos** of daily chess activities conducted in the respective district through bootcamps & Platform Usage Pics. 2.A **dedicated section** will be provided to upload: • Activity images • Activity videos 3.A separate **district-wise achievement module** will be available to record and showcase students' achievements. 4.A **bootcamp attendance module** will be included to track and manage student attendance on a daily basis. 5.Student Remark section created for marking coach remarks on particular students during bootcamp.(This is to be collected from the coach and feeded by the manager-Chessbishop Internal)	It's crucial to the staff attendance and progress monitoring.This serves to be the proof of activity and strong document for district renewals.Hence required in phase in urgent basis.

2.2 Voicebot Refinement - Tamil & English

The current voicebot listens and responds in Tamil and English, but the accuracy and responsiveness need significant improvement. This is a Phase 1 priority because a slow or inaccurate voicebot is immediately noticeable during any demo or vetting session.

- **Latency target:** Under 500 milliseconds from speech end to response start.
- **Accuracy target:** 95%+ word recognition for chess-related terminology in Tamil and English.
- **Integration points:** Assessment instructions, puzzle hints, Mantri advisor, analysis narration.
- **Technology:** Sarvam AI for Tamil (best-in-class for Indic languages), ElevenLabs or Azure for English. Vishnu team evaluates and recommends.
- **Hindi voicebot:** Not in Phase 1. Planned for Phase 3 when state-deal discussions extend beyond Tamil Nadu.

2.3 Student Onboarding System

This is the operational deliverable that makes everything else usable.

- Bulk CSV import for district-wise student data.
- Auto-generation of student login credentials (Student ID + password).
- Per-district and per-school grouping so we can report properly later.
- Print-ready credential sheets coaches can distribute in schools.
- Existing 1,048 students imported cleanly without disruption.
- Options to Modify names for spelling correction/profile details additions by Admin.

2.4 Renewal Report Generator

A simple but critical piece: one-click generation of professional PDF reports per student, per school, and per district. Platform usage, progress trends, assessment results, improvement metrics. This is what justifies work-order renewals when we sit across from district officers.

2.5 "Coming Soon" Feature Shelf

All features planned for Phase 2 and Phase 3 will appear in the UI as clickable cards labelled “Coming Soon.” This is aesthetic but strategic:

- Campaign Map (Coming Soon)
- Interactive Video Lessons (Coming Soon)
- Coach Dashboard (Coming Soon)
- Virtual Live Classes (Coming Soon)
- Endgame Lab (Coming Soon)
- Middlegame Lab (Coming Soon)
- Puzzle Rush + Tournament Mode (Coming Soon)

When state vetting teams or new district officers see the platform, they will see the full vision on screen - not just what we shipped this month. This creates confidence and shows momentum. Low engineering cost, high perception value.

2.6 Video Lesson Infrastructure Foundation (Included in Phase 1)

My decision: include the video lesson player framework foundation in Phase 1, even though the actual videos and full interactive integration happen in Phase 2. Here is why:

- The framework is relatively quick to build (2–3 days of dev work).
- I will be producing Heygen + Higgsfield + Remotion videos in parallel starting this month.

- When Phase 2 arrives, video ingestion is plug-and-play instead of starting from scratch.
- The state team will see “Video Lessons” working (even with just 2–3 sample videos) rather than as another “Coming Soon” card.

This is a small investment in Phase 1 that gives us a visible, working feature during vetting instead of a promise.

2.7 What Is Not in Phase 1 (Deferred to Phase 2 or 3)

- Full Campaign Map with 23 levels
- Full gamification (XP, streaks, hearts, badges, leaderboards)
- Endgame Lab and Middlegame Position Lab content
- Coach Dashboard with homework system
- Virtual Live Classes (separate premium module - see Section 7)
- LLM position explanations for mistakes and blunders
- WC-ACPL engine (weighted contextual analysis)
- Tournament manager
- Scoresheet OCR scanner
- Hindi voicebot

3. Mobile-First Design - Critical Detail

This deserves its own section because it is easy to under-deliver on and hard to fix later.

3.1 Why Mobile-First Matters for Sigaram64

- Government school students primarily use smartphones at home.
- School computer labs are used for 30–45 minutes at a time; real practice happens on phones.
- State vetting teams will test the platform on their own phones during demos.
- Parents monitor student progress from phones, not laptops.
- Public launch (future phase) will see 80%+ traffic from mobile devices.

3.2 Mobile Design Requirements

Area	Mobile Requirement
Chess board	Full-width on mobile. Drag-to-move and tap-to-move both supported. Pieces large enough for finger accuracy (minimum 40px per square on a 5.5-inch screen).
Puzzle screen	Puzzle instruction, board, and controls visible without scrolling on standard mobile screens.
Assessment flow	Progress bar visible. Question area swipeable. No keyboard input required anywhere.
Navigation	Bottom tab bar (mobile standard) with 4–5 primary sections. Hamburger menu for secondary items.
Admin dashboard	Adaptive layout: stacked cards on mobile, grid on desktop. All tables scrollable horizontally.
Analysis screen	Board + move list + evaluation chart arranged vertically on mobile. Tabbed interface to switch between moves, chart, and commentary.
Typography	Base font size 16px minimum on mobile (no smaller). Tamil font renders cleanly at all sizes.
Touch targets	Minimum 44x44 pixel tap area for all buttons and interactive elements (WCAG + Apple/Google guidelines).
Performance	Pages load under 3 seconds on 4G. Puzzle board renders instantly. No layout shift after load.
Offline resilience	Assessment and puzzle play continue working if internet drops briefly. Syncs when connection returns.

3.3 Testing Commitment

All Phase 1 screens tested on at least 3 mobile devices before acceptance:

- Low-end Android (Redmi 9 or equivalent) - represents most government school students
- Mid-range Android (OnePlus/Samsung mid-tier)
- iPhone (latest iOS - represents state officials and vetting team members)

ℹ NOTE

If the platform looks polished on a Redmi 9 with mid-speed internet, it will look excellent on every other device. That is the quality bar I want us to hit.

4. Timeline

Work begins Monday, April 20, 2026. Platform goes live by May 30. Student warm-up completes in the first week of June. Stabilisation runs through the second week of June. Full data collection starts from June 15 onwards - aligned with the academic year.

4.1 Sprint Phases

Week	Dates	Focus	My Commitments
Week 1	Apr 20 - Apr 26	Kickoff meeting. Architecture and design review. Team spins up infrastructure. Initial UI mockups.	Attend kickoff. Approve architecture within 48 hours. Review mockups.
Week 2	Apr 27 - May 3	UI implementation starts (mobile-first). CAT assessment engine build. Event tracking SDK. Start voicebot refinement.	Sample district student data ready. Weekly demo on Friday May 1.
Week 3	May 4 - May 10	UI completion. Stockfish 18 integration. CPL classification. Glicko-2 system.	Validate analysis on 10 sample games. Content validation for assessment puzzles.
Week 4	May 11 - May 17	Admin dashboard v1. Bulk import system. Voicebot accuracy testing. Continued polish on UI.	Full district-wise student data delivered by May 15.
Week 5	May 18 - May 24	Renewal report generator. Admin dashboard final. End-to-end testing on mobile devices. Bug fixes.	Daily QA review. Feedback on report format.
Week 6	May 25 - May 30	Final polish. Production deployment. Pre-launch checks. Credential generation for all districts.	Final acceptance. Production sign-off by May 30.
Week 7 (Warm-up)	Jun 1 - Jun 7	Student credential distribution via coaches as schools reopen. Answer student queries. Monitor for any issues.	Coordinate with coaches. On-ground support. Collect early feedback.
Week 8 (Stabilisation)	Jun 8 - Jun 14	Platform stabilisation. Minor bug fixes from real-world usage. Performance tuning under real load.	Daily check-ins with team. Flag any issues immediately.
From June 15	-	Full data collection begins. Students using platform for academic year 2026-27.	Monitor dashboards. Begin producing first renewal reports by end of June.

4.2 Rhythm We Keep Together

- Daily standup on WhatsApp or Meet, 10 minutes, 9:30 AM Monday–Friday.
- Weekly Friday demo of working code, 30–45 minutes.
- Weekly Sunday written status update covering what shipped, what is blocked, and next week’s plan.
- Anything urgent - just call. The old way has always worked for us.

4.3 Honest Risk Discussion

What Could Slow Us Down	How We Handle It
Stockfish 18 NNUE integration proves tricky	Fall back to Stockfish 16 NNUE (mature, same architecture) without delaying timeline. Upgrade to 18 in Phase 2 if needed.
I delay UI approvals or student data delivery	I commit to 48-hour approval turnaround and May 15 deadline for student data. If I slip, timeline slips with me - fair to both sides.
Voicebot accuracy takes longer than planned	Ship Phase 1 with current voicebot improved to 90% accuracy target. Push to 95%+ in early Phase 2 if needed.
Bugs discovered during first-week warm-up in June	Week 8 (stabilisation week, June 8-14) exists specifically to absorb this. Team on standby.
Scope creep (new feature requests mid-sprint)	Anything new gets logged as Phase 2 wishlist. No mid-sprint additions unless we both agree they are small and essential.

ℹ NOTE
The six-week build window and two weeks of June buffer mean we are not rushing anything. June 15 is our real data-collection start date. Everything before that is preparation, not pressure.

6. Working Together

6.1 What I Bring

- Timely advance payment (within 10 working days of signing – that's a Max time).
- Architecture and UI sign-offs within 48 hours.
- District-wise student data by May 15 (Week 4, firm).
- 2 hours daily availability during Weeks 1–3 for chess content validation.
- Attendance at every Friday demo, no exceptions.
- Clear decisions when asked - no flip-flopping mid-sprint.
- Honest conversation if anything feels off, early rather than late.

6.2 What I Ask From the Team

- Named team members for each role (so we know who is working on what).
- Use of AI coding tools to maintain quality and velocity - this is modern best practice now.
- Daily standups and Friday demos without exception.
- Source code committed to a Sigaram64-owned Git repo throughout the sprint (I have read access from Day 1).
- Honest flagging of issues the moment they appear, not at the end of the week.
- Active presence during the first two weeks of June for stabilisation - real-world usage will surface things no amount of testing catches.
- A polished, mobile-first output we can both be proud of when state vetting happens.

6.3 Intellectual Property and Confidentiality

- All custom code, designs, and assets belong to Sigaram64 on final payment - standard terms, same as always between us.
- Mutual confidentiality on proprietary algorithms, business strategies, and user data.
- Student data remains on Sigaram64-controlled infrastructure throughout and after the engagement.

6.4 Revisions and Changes

- Up to 3 revision rounds per UI screen included - more than enough for clean work.
- Genuine mid-sprint change requests (not scope creep) handled with a quick written note. We have always worked well informally, and I want to keep that.
- Anything beyond normal back-and-forth goes into a formal change order with time and cost estimate.

7. Virtual Live Classes - A Separate Premium Module

This is worth its own section because it is our strongest lever for the Tamil Nadu state deal.

7.1 Why This Matters for SDAT / State Deal

The Tamil Nadu state team (SDAT and affiliated authorities) want to scale chess to the maximum number of students. Their preferred delivery model is not individual students on separate computers - it is one coach teaching 30–50 students together in a smart classroom. This is how most government-scale education programmes are actually delivered.

The platform that wins this deal will not be the one with the best individual puzzle engine. It will be the one that lets a single coach run an effective live session for many students at once, with automated tracking and reporting that proves the session actually happened and produced learning.

This is where we build something that beats Zoom and Google Meet decisively.

7.2 What a Purpose-Built Chess Live Class Looks Like

- **Coach hosts the class inside Sigaram64:** Students join via their Sigaram64 ID. Attendance captured automatically, no manual registers.
- **Synchronised demo board:** What the coach shows on their board appears on every student's screen in real time. Coach can move pieces and annotate.
- **Session recording:** Every class recorded automatically. Students can rewatch later inside the platform. Recordings tied to attendance records.
- **Live student mini-boards:** Each student has their own practice board beside the demo board. Coach can assign mid-class: "All students, find the best move in the next 30 seconds."
- **AI-tracked chess communication:** This is the differentiator. AI listens to the coach's audio and detects when they call out notation ("Knight to f6", "Castle kingside"). Matches spoken moves against the board position. Produces a transcript of every chess move and concept discussed in the session.
- **Engagement tracking:** Which students had their board visible during the session. Who attempted the mid-class exercises. Who stayed to the end. All reported to the coach dashboard.
- **Post-class puzzle packet:** System automatically generates 5–10 practice puzzles based on the themes the coach covered in the session. Students do these as homework - extending the classroom impact.

7.3 Why This Differentiates From Zoom/Google Meet

Capability	Zoom/Meet	Sigaram64 Live Classes
Attendance tracking	Manual or export CSV	Automatic via student login
Session recording	Yes (generic video)	Yes + mapped to student records + searchable by move
Demo board sync	Screen share (static, laggy)	Native synced chess board with coach controls
Student practice boards	No	Yes, each student has their own
Chess move detection	No	AI transcribes spoken notation to moves
Engagement metrics	Generic	Chess-specific: who solved what, when
Post-class homework	No	Auto-generated puzzles from session themes
Cost per student per month	Variable licensing	Included in Sigaram64 subscription

7.4 Pricing Discussion (Separate Module)

This module is not priced into Phase 1, Phase 2, or Phase 3. It is a dedicated premium feature we should build and price independently because:

- It is a separate cost centre (100ms API or similar has per-minute infrastructure costs).
- It is sold primarily to government/institutional clients at premium tiers, not individual public users.
- It has real recurring infrastructure cost, so the pricing model must reflect that.

I would like the team to prepare a separate module proposal with pricing tiers: base tier (live class + recording), standard tier (+ AI chess communication tracking), premium tier (+ full engagement analytics + auto-homework). We can present this to SDAT directly as a premium add-on when state discussions progress.

> **ℹ NOTE**
> This module is our SDAT differentiator. Please let us treat it as a focused separate build when the state deal moves to formal proposal stage. For now, just a “Coming Soon” card in the UI with the right imagery.

8. The Road Ahead - Phase 2 and Phase 3

I want to share the direction we are heading, even though I am only formally committing to Phase 1 today.

8.1 Phase 2 (Roughly July - August 2026)

The engagement layer of the platform:

- Campaign Map with 23 levels (Duolingo-style progression)
- XP, streaks, hearts, daily goals, badges, avatar customisation
- Endgame Lab (E1–E5: basic checkmates through rook endings)
- Middlegame Position Lab (3 core themes)
- Interactive Video Player full integration (my Heygen videos ingested)
- Coach Dashboard and homework system
- School/class leaderboards
- Per-school monthly PDF reports auto-generated and emailed

8.2 Phase 3 (Roughly September - October 2026)

The intelligence layer:

- LLM-powered position explanations for mistakes and blunders (Bloom-calibrated)
- WC-ACPL engine (weighted contextual centipawn analysis)
- Pattern detection across games (“You missed back-rank tactics 3 times this week”)
- Tournament manager (Swiss, Arena, Round Robin)
- Advanced endgame positions E6–E10
- Hindi voicebot
- Mantri AI advisor refinement

8.3 Beyond (Future Phases)

- Virtual Live Classes module (Section 7 - separate premium build, likely alongside SDAT deal)
- Public launch infrastructure (freemium, billing, rewards, referrals)
- AI Avatar video production pipeline (Heygen + Higgsfield + Remotion - I run this myself)
- AI agent for PGN-to-script extraction (I build this for content scaling)
- Scoresheet OCR scanner
- Native mobile app (iOS/Android)
- RL-based adaptive engine (after we have 3–6 months of event data)

We will plan and price Phase 2 and Phase 3 together after Phase 1 is delivered and we have caught our breath. The indicative order-of-magnitude is ₹8–9 lakh for Phase 2 and ₹5–6 lakh for Phase 3, but we will refine this based on what we actually learn during Phase 1.

9. Let Us Build This

Phase 1 is a focused, fair, achievable sprint. ₹12.86 lakh. Six weeks of careful build (April 20 - May 30), one week of warm-up (first week of June), one week of stabilisation (second week of June). Real data collection begins from June 15 onwards, aligned with the academic year.

Clear scope, mobile-first quality, voicebot that actually works, analysis that matches international standards. All ready when students walk back into their classrooms in June.

If you agree with the spirit of this document and want to adjust specifics, let us get on a call this week and finalise before Monday. I can sign, transfer the advance, and we begin fresh on Monday April 20.

Thank you again for everything you have done for Sigaram64 so far. The next year is going to be a good one for all of us.

Warm regards,

Professor SuryaKumar S A

Founder, Sigaram64 • FIDE National Instructor

Acknowledged and Accepted:

For Sigaram64:

Professor SuryaKumar S A Signature: ________________________ Date: ____________

For Allreal Solutions:

Vishnu T U Signature: ________________________ Date: ____________

Built together, step by step.